

Bashar Mqasqas

mqasqasbashar@gmail.com | github.com/mqasqas | linkedin.com/in/mqasqas

EDUCATION

Michigan State University, College of Engineering – East Lansing, MI

GPA: 3.93

Bachelor of Science, Electrical Engineering

August 2024 – Present

- Relevant Coursework: Circuits, Electromagnetic Fields and Waves, Microprocessors, Digital Logic, Programming in C and Python, Differential Equations
- Dean's List in all eligible semesters
- Expected graduation date: May 2028

EXPERIENCE

Visteon Corporation – Van Buren Township, MI

June 2026 – August 2026

Optics and OLED Display Intern

- Configured a Keysight DAQ970A system to run a luminance degradation life test across 2 suppliers, 3 thermal conditions (6 OLED displays total), and 4 greyscale drive levels for each display
- Prevented reversed light sensor polarity and a miswired connection from causing failures across 6 2-layer measurement boards by isolating both defects on a one-off board through oscilloscope probing
- Developed a chemical-kinetics-based predictive model in MATLAB, replicating results from a standard 10,000-hour test length with ~1,500 hours of data collection
- Validated a 1 kHz to 1 MHz frequency response on an extended under-panel camera (UPC) transimpedance amplifier design through TINA simulation and bench measurement using a function generator and oscilloscope

MSU Rocketry – East Lansing, MI

September 2024 – Present

PCB Team Member

- Engineered an accelerometer PCB using KiCad as part of an electrical system collecting flight data for student-built rockets, capturing a measured apogee of 9,964 ft in flight
- Integrated the PCB into flight systems by coordinating with payload and electronics bay teams, achieving system-level readiness ahead of the International Rocket Engineering Competition (IREC)
- Secured a 21st-place finish out of 87 teams in build quality at IREC

PROJECTS

Altoids Tin Headphone Amplifier

- Assembled a PCB-based op-amp audio amplifier from schematic documentation by hand soldering through-hole components, including resistors, capacitors, IC socket, and audio interfaces
- Characterized amplifier performance using a function generator (1 kHz, 200 mV peak-to-peak sine wave) and oscilloscope, confirming a 5x voltage gain at full volume with no DC offset present

USB-C Modification for DualShock 4 Controller

- Replaced the stock micro-USB breakout board with a custom USB-C counterpart to improve charging speed and connection reliability, validating functionality through multimeter testing and hands-on use
- Populated a custom USB-C PCB via solder paste and reflow soldering, sourcing components from open-source design files

Go-Kart Fabrication

- Led a 3-person team through frame, engine, and axle assembly to build a go-kart supporting full rider weight at speeds of 35+ MPH

SKILLS

- PCB Design (ECAD) and Assembly: KiCad, reflow soldering, hand soldering
- Measurements and Instrumentation: Oscilloscope and multimeter testing
- Programming/Simulation: C, Python, MATLAB, Verilog, PSpice, TINA
- Hardware: PC assembly, hardware troubleshooting
- Microsoft Office (Excel, Word, PowerPoint)